



one guiding centre, 150 collisions:
each kick moves it about one gyro-radius ρ

the estimate that follows

$N = \nu t$ collisions in time t

$$\langle \Delta r^2 \rangle \sim N \rho^2 = \nu t \rho^2$$

$$D_{\text{cl}} \sim \rho^2 \nu / 2$$

a diffusion coefficient built from ONE step size and
ONE step rate --- this is the CLASSICAL estimate

Real tokamaks lose heat faster than this. Toroidal geometry
adds a neoclassical enhancement, and TURBULENCE adds a
further channel that this picture does not model at all.

Explanatory sketch; schematic, not to scale.